

```
!pip install yfinance
```

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Requirement already satisfied: yfinance in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: pandas>=1.3.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: numpy>=1.16.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: requests>=2.31 in /usr/local/lib/python3.11/dist-packages
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Requirement already satisfied: peewee>=3.16.2 in /usr/local/lib/python3.11/dist-packages
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Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.11/dist-packages
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Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages

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```
!pip install torch torchvision torchaudio
```

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Requirement already satisfied: torch in /usr/local/lib/python3.11/dist-packages (2.3.0)
Requirement already satisfied: torchvision in /usr/local/lib/python3.11/dist-packages (0.18.0)
Requirement already satisfied: torchaudio in /usr/local/lib/python3.11/dist-packages (2.3.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.11/dist-packages (3.12.2)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.11/dist-packages (4.10.0)
Requirement already satisfied: networkx in /usr/local/lib/python3.11/dist-packages (3.2.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.11/dist-packages (3.1.2)
Requirement already satisfied: fsspec in /usr/local/lib/python3.11/dist-packages (2024.9.0)
Collecting nvidia-cuda-nvrtc-cu12==12.4.127 (from torch)
  Downloading nvidia_cuda_nvrtc_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (1.0 MB)
Collecting nvidia-cuda-runtime-cu12==12.4.127 (from torch)
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Collecting nvidia-cuda-cupti-cu12==12.4.127 (from torch)
  Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (1.0 MB)
Collecting nvidia-cudnn-cu12==9.1.0.70 (from torch)
  Downloading nvidia_cudnn_cu12-9.1.0.70-py3-none-manylinux2014_x86_64.whl (693.2 MB)
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Collecting nvidia-cufft-cu12==11.2.1.3 (from torch)
  Downloading nvidia_cufft_cu12-11.2.1.3-py3-none-manylinux2014_x86_64.whl (161.4 MB)
Collecting nvidia-curand-cu12==10.3.5.147 (from torch)
  Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl (165.4 MB)
Collecting nvidia-cusolver-cu12==11.6.1.9 (from torch)
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  Downloading nvidia_nccl_cu12-2.21.5-py3-none-manylinux2014_x86_64.whl (192.4 MB)
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Collecting nvidia-nvtx-cu12==12.4.127 (from torch)
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Requirement already satisfied: nvidia-cusparse-cu12==12.3.1.170 (from torch)
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Requirement already satisfied: sympy==1.13.1 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: numpy in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: pillow!=8.3.*,>=5.3.0 in /usr/local/lib/python3.11/dist-packages
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages
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Downloading nvidia_cuda_cupti_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (13.8 MB)
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Downloading nvidia_curand_cu12-10.3.5.147-py3-none-manylinux2014_x86_64.whl (56.3 MB)
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Downloading nvidia_cusparses_cu12-12.3.1.170-py3-none-manylinux2014_x86_64.whl (207.5 MB)
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Downloading nvidia_nvjitlink_cu12-12.4.127-py3-none-manylinux2014_x86_64.whl (21.1 MB)
_____ 21.1/21.1 MB 80.2 MB/s eta 0:00:00
Installing collected packages: nvidia-nvjitlink-cu12, nvidia-curand-cu12, nvidia-c

```

```

import yfinance as yf
import pandas as pd
import numpy as np
import torch
import torch.nn as nn
import torch.optim as optim
import random
from collections import deque

```

```

symbol = "AAPL"
start_date = "2020-01-01"
end_date = "2025-03-18"

```

```
data = yf.download(symbol, start=start_date, end=end_date)
```



```

YF.download() has changed argument auto_adjust default to True
[*****100%*****] 1 of 1 completed

```

```

data['SMA_5'] = data['Close'].rolling(window=5).mean()
data['SMA_20'] = data['Close'].rolling(window=20).mean()
data['Returns'] = data['Close'].pct_change()

```

```

data.dropna(inplace=True)
data.reset_index(drop=True, inplace=True)

```

```
Actions = {0: "HOLD", 1: "BUY", 2: "SELL"}
```

```

def get_state(data, index):
    return np.array([
        float(data.loc[index, 'Close']),
        float(data.loc[index, 'SMA_5']),
        float(data.loc[index, 'SMA_20']),
        float(data.loc[index, 'Returns'])
    ])

class TradingEnvironment:
    def __init__(self, data):
        self.data = data
        self.initial_balance = 10000
        self.balance = self.initial_balance
        self.holdings = 0
        self.index = 0

    def reset(self):
        self.balance = self.initial_balance
        self.holdings = 0
        self.index = 0
        return get_state(self.data, self.index)

    def step(self, action): # Indented step function to make it a method of TradingEnv
        price = float(self.data.loc[self.index, 'Close'])
        reward = 0

        if action == 1 and self.balance >= price: # BUY
            self.holdings = self.balance // price # Correct the typo and update holdi
            self.balance -= self.holdings * price
        elif action == 2 and self.holdings > 0: # SELL
            self.balance += self.holdings * price
            self.holdings = 0

        self.index += 1
        done = self.index >= len(self.data) - 1

        # Calculate reward based on portfolio value change
        current_portfolio_value = self.balance + self.holdings * price
        reward = current_portfolio_value - (self.initial_balance if self.index == 1 el

        if done:
            reward = self.balance - self.initial_balance # Final reward at the end

        next_state = get_state(self.data, self.index) if not done else None
        return next_state, reward, done, {}

class DQN(nn.Module):
    def __init__(self, state_size, action_size):
        super(DQN, self).__init__()
        self.fc1 = nn.Linear(state_size, 64)
        self.fc2 = nn.Linear(64, 64)

```

```
self.fc2 = nn.Linear(64, 64)
self.fc3 = nn.Linear(64, action_size)
```

```
def forward(self, x):
    x = torch.relu(self.fc1(x))
    x = torch.relu(self.fc2(x))
    return self.fc3(x)
```

```
class DQNAgent:
```

```
    def __init__(self, state_size, action_size):
        self.state_size = state_size
        self.action_size = action_size
        self.memory = deque(maxlen = 2000)
        self.gamma = 0.95    #Discount factor
        self.epsilon = 1.0   #Exploration Rate
        self.epsilon_min = 0.01
        self.epsilon_decay = 0.995
        self.learning_rate = 0.001
        self.model = DQN(state_size, action_size)
        self.optimizer = optim.Adam(self.model.parameters(), lr=self.learning_rate)
        self.criterion = nn.MSELoss()

    def remember(self, state, action, reward, next_state, done):
        self.memory.append((state, action, reward, next_state, done))

    def act(self, state):
        if random.uniform(0,1) < self.epsilon:
            # Use 'Actions' instead of 'ACTIONS'
            return random.choice(list(Actions.keys()))
        state = torch.FloatTensor(state).unsqueeze(0)
        with torch.no_grad():
            q_values = self.model(state)
        return torch.argmax(q_values).item()

    def replay(self, batch_size):
        if len(self.memory) < batch_size:
            return
        minibatch = random.sample(self.memory, batch_size)
        for state, action, reward, next_state, done in minibatch:
            target = reward
            if not done:
                # Convert next_state to a PyTorch tensor before unsqueezing
                next_state_tensor = torch.FloatTensor(next_state).unsqueeze(0)
                target += self.gamma*torch.max(self.model(next_state_tensor)).item()

            state_tensor = torch.FloatTensor(state).unsqueeze(0)
            target_tensor = self.model(state_tensor).clone().detach()
            target_tensor[0][action] = target

            self.optimizer.zero_grad()
            output = self.model(state_tensor)
            loss = self.criterion(output, target_tensor)
```

```

        loss.backward()
        self.optimizer.step()

    if self.epsilon > self.epsilon_min:
        self.epsilon *= self.epsilon_decay

env = TradingEnvironment(data)
agent = DQNAgent(state_size = 4, action_size = 3)
batch_size = 32
episodes = 500
total_rewards = []

for episode in range(episodes):

    state = env.reset()
    done = False
    total_reward = 0

    while not done:
        action = agent.act(state)
        next_state, reward, done, _ = env.step(action)
        agent.remember(state, action, reward, next_state, done)
        state = next_state
        total_reward += reward

    agent.replay(batch_size)
    total_rewards.append(total_reward)
    print(f"Episodes {episode+1}/{episodes}, Total Reward: {total_reward}")

print("Training Complete!")

```

```

<ipython-input-8-3e6e3d58a2f9>:3: FutureWarning: Calling float on a single element
  float(data.loc[index, 'Close']),
<ipython-input-8-3e6e3d58a2f9>:4: FutureWarning: Calling float on a single element
  float(data.loc[index, 'SMA_5']),
<ipython-input-8-3e6e3d58a2f9>:5: FutureWarning: Calling float on a single element
  float(data.loc[index, 'SMA_20']),
<ipython-input-8-3e6e3d58a2f9>:6: FutureWarning: Calling float on a single element
  float(data.loc[index, 'Returns']),
<ipython-input-9-0e40d2ca55b7>:16: FutureWarning: Calling float on a single elemen
  price = float(self.data.loc[self.index, 'Close'])
<ipython-input-9-0e40d2ca55b7>:31: FutureWarning: Calling float on a single elemen
  reward = current_portfolio_value - (self.initial_balance if self.index == 1 else
Episodes 1/500, Total Reward: -9866.82787322998
Episodes 2/500, Total Reward: -9922.322605133057
Episodes 3/500, Total Reward: -9942.201251983643
Episodes 4/500, Total Reward: -9897.570693969727
Episodes 5/500, Total Reward: 17875.68310546875
Episodes 6/500, Total Reward: -9797.07292175293
Episodes 7/500, Total Reward: 9865.532600402832
Episodes 8/500, Total Reward: -9852.049034118652
Episodes 9/500, Total Reward: -9838.157318115234
Episodes 10/500, Total Reward: -9994.32901763916

```

```

Episodes 11/500, Total Reward: -9860.949348449707
Episodes 12/500, Total Reward: -9924.787300109863
Episodes 13/500, Total Reward: -9869.346736907959
Episodes 14/500, Total Reward: -9806.022750854492
Episodes 15/500, Total Reward: -9843.432258605957
Episodes 16/500, Total Reward: -9945.844139099121
Episodes 17/500, Total Reward: -9818.899719238281
Episodes 18/500, Total Reward: -9865.576641082764
Episodes 19/500, Total Reward: -9833.282665252686
Episodes 20/500, Total Reward: -9877.149105072021
Episodes 21/500, Total Reward: -1251.1103172302246
Episodes 22/500, Total Reward: -9828.199367523193
Episodes 23/500, Total Reward: -9861.128353118896
Episodes 24/500, Total Reward: -9937.075908660889
Episodes 25/500, Total Reward: -9808.23430633545
Episodes 26/500, Total Reward: -9894.521934509277
Episodes 27/500, Total Reward: -9854.764766693115
Episodes 28/500, Total Reward: -9946.928462982178
Episodes 29/500, Total Reward: -9842.41138458252
Episodes 30/500, Total Reward: -9856.779594421387
Episodes 31/500, Total Reward: -9845.121963500977
Episodes 32/500, Total Reward: -9759.83967590332
Episodes 33/500, Total Reward: -9882.86771774292
Episodes 34/500, Total Reward: -9771.976612091064
Episodes 35/500, Total Reward: -9881.76411819458
Episodes 36/500, Total Reward: -9857.822856903076
Episodes 37/500, Total Reward: -9923.676315307617
Episodes 38/500, Total Reward: -9803.502174377441
Episodes 39/500, Total Reward: 6753.141151428223
Episodes 40/500, Total Reward: 12089.568920135498
Episodes 41/500, Total Reward: 10034.724304199219
Episodes 42/500, Total Reward: 21154.55407333374
Episodes 43/500, Total Reward: 13567.808521270752
Episodes 44/500, Total Reward: 4708.2394371032715
Episodes 45/500, Total Reward: -9869.343460083008
Episodes 46/500, Total Reward: -9929.34228515625

```

```

test_env = TradingEnvironment(data)
state = test_env.reset()
done = False

while not done:
    action = agent.act(state)
    next_state, reward, done, _ = test_env.step(action)
    state = next_state if next_state is not None else state

final_balance = test_env.balance
profit = final_balance - test_env.initial_balance
print(f"Final Balance after testing: ${final_balance:.2f}")
print(f"Total Profit: ${profit:.2f}")

```

```

<ipython-input-8-3e6e3d58a2f9>:3: FutureWarning: Calling float on a single element
float(data.loc[index, 'Close']),
<ipython-input-8-3e6e3d58a2f9>:4: FutureWarning: Calling float on a single element
float(data.loc[index, 'SMA_5']),
<ipython-input-8-3e6e3d58a2f9>:5: FutureWarning: Calling float on a single element
float(data.loc[index, 'SMA_20']),
<ipython-input-8-3e6e3d58a2f9>:6: FutureWarning: Calling float on a single element

```

```
<ipython-input-9-3ee3a3ca2197>: FutureWarning: Calling float on a single element
  float(data.loc[index, 'Returns']),
<ipython-input-9-0e40d2ca55b7>:16: FutureWarning: Calling float on a single elemen
  price = float(self.data.loc[self.index, 'Close'])
<ipython-input-9-0e40d2ca55b7>:31: FutureWarning: Calling float on a single elemen
  reward = current_portfolio_value - (self.initial_balance if self.index == 1 else
Final Balance after testing: $25133.74
Total Profit: $15133.74
```

Start coding or [generate](#) with AI.